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DS Homework 2:-

QUES 1: Define an integer array. Declare an integer pointer and initialize it to point to first element of array. Use pointer arithmetic to traverse the array and print the elements. Modify the elements of the array using pointer arithmetic.

```
#include<stdio.h>
void printArray(int *ptr, int n) {
   for (int i = 0; i < n; i++) {
      printf("%d ", *(ptr + i));
   }
   printf("\n");
}
int main() {
   int n;
   printf("Enter the size of Array\n");
   scanf("%d",&n);
   int arr[n];
   printf("Enter array elements\n");
   for(int i=0;i<n;i++)
   scanf("%d",&arr[i]);
   int *ptr = arr;
   printf("Original array: ");
   printArray(ptr, n);
   for (int i = 0; i < n; i++) {
      *(ptr + i) = *(ptr + i) * 2;
   }
   printf("Modified array (doubled): ");
   printArray(ptr, n);
   return 0;
}
```

Output:-

```
Enter the size of Array
4
Enter array elements
1
2
3
4
Original array: 1 2 3 4
Modified array (doubled): 2 4 6 8
```

QUES 2: Calculate string length using pointer.

```
#include <stdio.h>
#include <string.h>
int stringLength(char *str) {
   char *p = str;
   while (*p != '\0') {
      p++;
   }
   return (int)(p - str);
}
int main() {
   char str[100];
   printf("Enter a string: ");
   fgets(str, sizeof(str), stdin);
   size_t len = strlen(str);
   if (len > 0 && str[len - 1] == '\n') {
      str[len - 1] = '\0';
   }
   printf("Length of \"%s\" is %d\n", str, stringLength(str));
   return 0;
}
```

Output:-

```
Enter a string: Rameez Hamza
Length of "Rameez Hamza" is 12
```

QUES 3: Write a function that calculate both the sum and product of the two numbers. Return these two results using pointer passed as arguments to the function.

```
#include <stdio.h>
void calculate(int a, int b, int *sum, int *product) {
   *sum = a + b;
   *product = a * b;
}
int main() {
   int a, b, sum, product;
   printf("Enter two numbers: ");
   scanf("%d %d", &a, &b);
   calculate(a, b, &sum, &product);
   printf("Sum = %d\n", sum);
```

```
    printf("Product = %d\n", product);
    return 0;
}
```

Output:-

```
Enter two numbers: 5
6
Sum = 11
Product = 30
```

QUES 4: Create an array of structure uses dynamic memory allocation to allocate memory space and use a structure pointer with pointer arithmetic to iterate and access modify members.

```
#include <stdio.h>
#include <stdlib.h>
typedef struct {
    int id;
    char name[30];
    float marks;
} Student;
int main() {
    int n;
    printf("Enter number of students: ");
    scanf("%d", &n);
    Student *students = (Student *) malloc(n * sizeof(Student));
    if (students == NULL) {
        printf("Memory allocation failed\n");
        return 1;
    }
    Student *ptr = students;
    for (int i = 0; i < n; i++) {
        printf("Enter id for student %d: ", i + 1);
        scanf("%d", &(ptr + i)->id);
        printf("Enter name for student %d: ", i + 1);
        scanf("%s",(ptr + i)->name);
        printf("Enter marks for student %d: ", i + 1);
        scanf("%f",&(ptr + i)->marks);
    }
    for (int i = 0; i < n; i++) {
        (ptr + i)->marks += 5;
```

```
    }
    printf("\nStudent details (after bonus):\n");
    for (int i = 0; i < n; i++) {
        printf("ID: %d, Name: %s, Marks: %.2f\n",
               (ptr + i)->id, (ptr + i)->name, (ptr + i)->marks);
    }
    free(students);
    return 0;
}
```

Output:-

```
Enter number of students:
2
Enter id for student 1: 101
Enter name for student 1: Rohan
Enter marks for student 1: 95
Enter id for student 2: 102
Enter name for student 2: Raj
Enter marks for student 2: 85

Student details (after bonus):
ID: 101, Name: Rohan, Marks: 100.00
ID: 102, Name: Raj, Marks: 90.00
```

QUES 5: Define two functions with same signature, declare a function pointer and use the pointer to call both function.

```
#include <stdio.h>
int add(int a, int b) {
    return a + b;
}
int multiply(int a, int b) {
    return a * b;
}
int main() {
    int (*funcPtr)(int, int);
    int a,b ;
    printf("Enter the two numbers\n");
    scanf("%d %d",&a,&b);
    funcPtr = add;
    printf("Sum = %d\n", funcPtr(a, b));
    funcPtr = multiply;
    printf("Product = %d\n", funcPtr(a, b));
}
```

Output:-

```
Enter the two numbers
4
5
Sum = 9
Product = 20
```

QUES 6: Declare an integer a pointer to it and a double pointer. Print the values and addresses at each level.

```
#include <stdio.h>
int main() {
   int num;
   printf("Enter the number\n");
   scanf("%d",&num);
   int *ptr = &num;
   int **dptr = &ptr;
   printf("Value of num: %d\n", num);
   printf("Address of num: %p\n", (void *)&num);
   printf("\nUsing pointer:\n");
   printf("Value of ptr (address of num): %p\n", (void *)ptr);
   printf("Value pointed by ptr (*ptr): %d\n", *ptr);
   printf("Address of ptr: %p\n", (void *)&ptr);
   printf("\nUsing double pointer:\n");
   printf("Value of dptr (address of ptr): %p\n", (void *)dptr);
   printf("Value pointed by dptr (*dptr, address of num): %p\n", (void *)*dptr);
   printf("Value pointed by *dptr (**dptr): %d\n", **dptr);
   printf("Address of dptr: %p\n", (void *)&&dptr);
}
```

Output:-

```
Enter the number
120
Value of num: 120
Address of num: 000000BF98BFFE5C

Using pointer:
Value of ptr (address of num): 000000BF98BFFE5C
Value pointed by ptr (*ptr): 120
Address of ptr: 000000BF98BFFE50

Using double pointer:
Value of dptr (address of ptr): 000000BF98BFFE50
Value pointed by dptr (*dptr, address of num): 000000BF98BFFE5C
Value pointed by *dptr (**dptr): 120
Address of dptr: 000000BF98BFFE48
```

QUES 7: Sort an array of floating point numbers using selection sort.

```
#include <stdio.h>
void selectionSort(float arr[], int n) {
   for (int i = 0; i < n - 1; i++) {
      int minIndex = i;
      for (int j = i + 1; j < n; j++) {
         if (arr[j] < arr[minIndex]) {
            minIndex = j;
         }
      }
      if (minIndex != i) {
         float temp = arr[i];
         arr[i] = arr[minIndex];
         arr[minIndex] = temp;
      }
   }
}
int main() {
   int n;
   printf("Enter number of elements: ");
   scanf("%d", &n);
   float arr[n];
   printf("Enter %d floating point numbers: ", n);
   for (int i = 0; i < n; i++) {
      scanf("%f", &arr[i]);
   }
   selectionSort(arr, n);
   printf("Sorted array: ");
   for (int i = 0; i < n; i++) {
      printf("%.2f ", arr[i]);
   }
   printf("\n");
   return 0;
}
```

Output:-

```
Enter number of elements: 3
Enter 3 floating point numbers: 2.05
2.25
2.26
Sorted array: 2.05 2.25 2.26
```

QUES 8: Define a structure student with id , name and grade create an array of student structures and sort them based on different feed.

```
#include <stdio.h>
#include <string.h>
typedef struct {
   int id;
   char name[30];
   float grade;
} Student;
void sortById(Student arr[], int n) {
   for (int i = 0; i < n - 1; i++)
      for (int j = 0; j < n - i - 1; j++)
         if (arr[j].id > arr[j + 1].id) {
            Student temp = arr[j];
            arr[j] = arr[j + 1];
            arr[j + 1] = temp;
         }
}
void sortByName(Student arr[], int n) {
   for (int i = 0; i < n - 1; i++)
      for (int j = 0; j < n - i - 1; j++)
         if (strcmp(arr[j].name, arr[j + 1].name) > 0) {
            Student temp = arr[j];
            arr[j] = arr[j + 1];
            arr[j + 1] = temp;
         }
}
void sortByGrade(Student arr[], int n) {
   for (int i = 0; i < n - 1; i++)
      for (int j = 0; j < n - i - 1; j++)
         if (arr[j].grade < arr[j + 1].grade) { /* descending order */
            Student temp = arr[j];
            arr[j] = arr[j + 1];
            arr[j + 1] = temp;
         }
}
void printStudents(Student arr[], int n) {
   for (int i = 0; i < n; i++) {
      printf("ID: %d, Name: %s, Grade: %.2f\n", arr[i].id, arr[i].name, arr[i].grade);
   }
   printf("\n");
}
```

```
int main() {
    int n;
    printf("Enter number of students: ");
    scanf("%d", &n);
    Student students[n];
    for (int i = 0; i < n; i++) {
        printf("Enter id for student %d: ", i + 1);
        scanf("%d", &students[i].id);
        printf("Enter name for student %d: ", i + 1);
        scanf("%s",students[i].name);
        printf("Enter grade for student %d: ", i + 1);
        scanf("%f",&students[i].grade);
    }
    sortById(students, n);
    printf("\nSorted by ID:\n");
    printStudents(students, n);

    sortByName(students, n);
    printf("Sorted by Name:\n");
    printStudents(students, n);

    sortByGrade(students, n);
    printf("Sorted by Grade (descending):\n");
    printStudents(students, n);
    return 0;
}
```

Output:-

```
Enter number of students: 3
Enter id for student 1: 101
Enter name for student 1: Raj
Enter grade for student 1: 85
Enter id for student 2: 102
Enter name for student 2: Rohan
Enter grade for student 2: 75
Enter id for student 3: 102
Enter name for student 3: Aman
Enter grade for student 3: 82

Sorted by ID:
ID: 101, Name: Raj, Grade: 85.00
ID: 102, Name: Rohan, Grade: 75.00
ID: 102, Name: Aman, Grade: 82.00

Sorted by Name:
ID: 102, Name: Aman, Grade: 82.00
ID: 101, Name: Raj, Grade: 85.00
ID: 102, Name: Rohan, Grade: 75.00

Sorted by Grade (descending):
ID: 101, Name: Raj, Grade: 85.00
ID: 102, Name: Aman, Grade: 82.00
ID: 102, Name: Rohan, Grade: 75.00
```

QUES 9: Sort a set of negative numbers using radix sort.

```
#include <stdio.h>
#include <stdlib.h>
int getMax(int arr[], int n) {
   int max = arr[0];
   for (int i = 1; i < n; i++)
      if (arr[i] > max) max = arr[i];
   return max;
}
void countingSort(int arr[], int n, int exp) {
   int *output = (int *) malloc(n * sizeof(int));
   int count[10] = {0};
   for (int i = 0; i < n; i++)
      count[(arr[i] / exp) % 10]++;

   for (int i = 1; i < 10; i++)
      count[i] += count[i - 1];

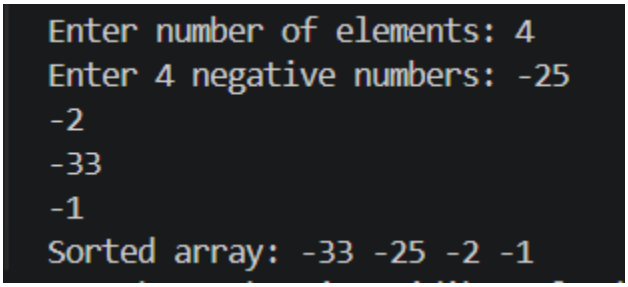
   for (int i = n - 1; i >= 0; i--) {
      int digit = (arr[i] / exp) % 10;
      output[count[digit] - 1] = arr[i];
      count[digit]--;
   }
   for (int i = 0; i < n; i++)
      arr[i] = output[i];
   free(output);
}
void radixSort(int arr[], int n) {
   if (n == 0) return;
   int max = getMax(arr, n);
   for (int exp = 1; max / exp > 0; exp *= 10)
      countingSort(arr, n, exp);
}
void radixSortNegatives(int arr[], int n) {
   int *absArr = (int *) malloc(n * sizeof(int));
   for (int i = 0; i < n; i++)
      absArr[i] = -arr[i];

   radixSort(absArr, n);
   for (int i = 0; i < n; i++)
      arr[i] = -absArr[n - 1 - i];

   free(absArr);
}
```

```
int main() {
   int n;
   printf("Enter number of elements: ");
   scanf("%d", &n);
   int *arr = (int *) malloc(n * sizeof(int));
   printf("Enter %d negative numbers: ", n);
   for (int i = 0; i < n; i++)
      scanf("%d", &arr[i]);
   radixSortNegatives(arr, n);
   printf("Sorted array: ");
   for (int i = 0; i < n; i++)
      printf("%d ", arr[i]);
   printf("\n");
   free(arr);
   return 0;
}
```

Output:-



QUES 10: Write a C program to copy one array to another using pointers.

```
#include <stdio.h>
void copyArray(int *src, int *dest, int n) {
   for (int i = 0; i < n; i++) {
      *(dest + i) = *(src + i);
   }
}
int main() {
   int n;
   printf("Enter number of elements: ");
   scanf("%d", &n);
   int source[n], destination[n];
   printf("Enter %d elements: ", n);
   for (int i = 0; i < n; i++)
      scanf("%d", &source[i]);
```

```
    copyArray(source, destination, n);
    printf("Copied array: ");
    for (int i = 0; i < n; i++)
        printf("%d ", destination[i]);
    printf("\n");
    return 0;
}
```

Output:-

```
Enter number of elements: 4
Enter 4 elements: 1
2
3
4
Copied array: 1 2 3 4
```